MODEL E

Research History, Design Evolution and Lessons Learned
Muon-decay neutrino communication: a documented negative feasibility study

Status: Exploratory, non-peer-reviewed, human–AI-assisted modeling. No experimental neutrino communication link was built or demonstrated. November 2026.

Executive summary

Model E began with a question: could neutrinos carry a low-bandwidth on/off message from a submerged platform through seawater to a receiver near the surface? The investigation progressed from speculative source concepts to a muon-decay transmitter, a 100-tonne water-Cherenkov receiver, numerical geometry studies and synthetic packet-decoding experiments. It ended because the estimated particle throughput and associated energy demand were far beyond plausible submarine engineering limits. The strongest result is a clearly documented feasibility obstacle, not a working communications design.

1. Origin: communication through seawater and solids

The motivating use case was communication with a submerged, potentially autonomous vessel, where conventional electromagnetic transmission is strongly constrained by seawater and acoustic links have different operational limitations. The early goal was deliberately modest: reliable binary signaling rather than high data rates. Neutrinos were attractive because they traverse matter with very little attenuation, but that same weak interaction makes detection exceptionally difficult.

2. Early source ideas and screening

Concept considered	Reason it was not retained as baseline
Modulating natural stellar neutrinos	No credible practical mechanism for controlling the ambient flux or flavor distribution as a transmitter.
Reactor-related neutrino production	A reactor is a source of neutrinos, but the investigated concept did not provide controllable, efficiently detected modulation.
High-energy photon / secondary particle production	Prompted examination of pion–muon channels; efficient, compact production was not established.
Gravitational lensing / exotic astrophysical effects	Not an engineering pathway for a controllable terrestrial link.
Muon decay with magnetic steering	Chosen as the tractable theoretical baseline, while acknowledging major production and directional losses.

3. Evolution of the transmitter

The working chain became: candidate tungsten target; short pion channel; muon selection and storage; timed magnetic control; muon decay to neutrinos. A nominal muon kinetic energy of 100 MeV and magnetic field of 0.2 T were used. For a 100 MeV kinetic-energy muon, the Lorentz factor is approximately 1.946 and beta approximately 0.858. A nominal 2 microsecond storage interval, roughly five turns in a selected extraction window (~359 ns), and a 30-degree useful arc were explored. These values were modeling choices, not a validated source design.

Alternatives included steady magnetic fields versus pulsed steering, rejecting unwanted muons, delayed extraction, and straight decay-channel re-injection. The orientation of the storage ring matters: a horizontal ring cannot direct tangents vertically toward a surface receiver, whereas a suitably oriented vertical ring can offer upward-going tangents over part of its circumference. Decay-point position, tangent direction and energy-angle correlations remain essential for any credible flux calculation.

4. Why directionality did not solve the problem

Relativistic decay favors the forward direction. The characteristic opening angle 1/gamma is about 29 degrees for the reference muons, but a broad forward cone is not a narrow beam. A quoted probability of emission into that cone is not the probability of striking a small receiver a kilometer away. The study therefore moved toward geometric integration over ring arcs and receiver volumes, while recognizing that simplified angular assumptions can substantially affect absolute event rates.

5. Receiver: water Cherenkov concept

The receiver baseline was 100 tonnes of water (approximately 100 cubic meters), placed on a surface ship directly above the submarine. A reference cylinder 2 m high has a diameter near 7.98 m. Alternative cylinder heights from 0.5 to 10 m were explored. In an optically thin, nearly uniform neutrino flux, changing area and depth while keeping volume fixed does not substantially change the number of neutrino interactions. However, realistic photon collection, trigger thresholds, event topology, background rejection, mechanical feasibility and track containment were not solved by this simple argument.

6. Numerical geometry exploration

Exploratory Python scans compared water-cylinder shapes, separation distances from 100 m to 5,000 m, and selected muon-ring arcs. These scripts are reproducible research artifacts but use a simplified interaction probability. In particular, a constant 10^-40 cm² reference cross section per free proton was used in parts of the modeling; this is not a validated energy- and flavor-averaged water-Cherenkov response. Muon neutrino, electron neutrino and antineutrino interaction channels must be handled separately in a physically calibrated model.

7. Transition to communications and signal processing

The signaling scheme used on–off keying (OOK), nominal 10 Hz event sampling, and one-second bits in the initial experiments. A packet contained a four-bit sync word 0101, eight payload bits and two CRC-2 bits (14 bits total). Synthetic background event streams included solar, cosmic and instrumental components and occasional bursts. Background estimators A1, A2 and A3 were compared, as were pairwise and three-way averages. A1 used the raw recent window; A2 replaced peaks assigned to verified packets; A3 also attempted to remove candidate packets that failed validation. A3 may suppress genuine unrelated background fluctuations and bias its own estimate.

A critical limitation: some experiments used known packet starts rather than fully blind continuous synchronization. The simulated event rates were chosen for algorithm testing, not derived from a physically validated muon-to-detector flux. Reported bit and packet error rates therefore cannot be used as evidence of an achievable real-world channel.

8. Shorter bits, sampling and information rate

For 14-bit packets with eight payload bits, the ideal net throughput before packet errors is 8/(14 T_bit). At 1.0 s/bit this is 0.571 bit/s; at 0.5 s, 1.143 bit/s; and at 0.2 s, 2.857 bit/s. Suggested sampling rates of roughly 10–50 Hz offer multiple time bins per bit, but finer binning does not create additional neutrino events or overcome Poisson counting statistics. At shorter bits, maintaining comparable counts per bit requires a higher detected event rate.

9. The decisive 1,000 m energy budget

An illustrative target of 13 detected signal events per bit was carried forward from synthetic tests. Using a simplified per-stored-muon detection probability of approximately 1.91 x 10^-21 at 1 km (including a 2 microsecond decay fraction and 30-degree arc fraction), the nominal 0.2 s/bit case would require about 3.4 x 10^22 stored muons per second. At 100 MeV kinetic energy per muon, this corresponds to approximately 545 GW of muon kinetic beam power alone. The value is a conditional toy-model scale estimate, not a validated physical prediction; target, acceleration, transport, magnet, cooling and electrical losses would increase the required input power.

Bit duration	Ideal net payload rate	Toy-model muon rate	Muon kinetic power
0.1 s	5.71 bit/s	6.8 x 10^22 /s	1.09 TW
0.2 s	2.86 bit/s	3.4 x 10^22 /s	545 GW
0.3 s	1.90 bit/s	2.3 x 10^22 /s	~363 GW
0.5 s	1.14 bit/s	1.36 x 10^22 /s	~218 GW

The table fixes 13 expected signal events per bit. It does not imply equal packet-error rates across durations, and it excludes the rest-mass energy budget and production inefficiency. Even if the toy-model absolute interaction rate changes after proper calibration, the required source power is a central feasibility obstacle.

10. Beyond the Standard Model: speculative closing assessment

Supersymmetry, string-inspired extra dimensions and non-standard neutrino interactions were discussed as possible future changes to production, propagation or detection. No experimentally established mechanism from these frameworks currently provides the many orders of magnitude improvement needed for this design. New interactions might increase some detection channels, but they could also affect propagation and face strong experimental constraints. These possibilities must not be presented as engineering solutions.

11. What failed, what was learned, and what remains open

Category	Assessment
Engineering feasibility	Failed: the investigated transmitter–receiver architecture requires extreme particle throughput and beam power.
Physics calibration	Incomplete: energy-dependent cross sections, neutrino flavors, full decay kinematics and real Cherenkov efficiency were not validated.
Receiver practicality	Incomplete: ship integration, photodetector coverage, background rejection and trigger efficiency require detailed study.
Communications validation	Incomplete: fully blind packet acquisition, realistic background measurements, and independent replication remain outstanding.
Useful research output	A traceable sequence of hypotheses, simplifying assumptions, numerical scans, code, synthetic decoding comparisons and a clearly stated negative conclusion.

12. Reproducibility and interpretation

The repository should preserve the exploratory Python programs and output tables alongside this history and the separate feasibility report. Reported numbers are not independent experimental measurements; some earlier calculations and simulation claims may require verification against the archived code. No physical prototype was built. No formal peer review was performed. The most defensible conclusion is that the specific Model E configuration is not suitable for practical deployment under the stated assumptions—not that all conceivable neutrino communication systems are impossible.

Suggested next steps only if revisited

A future revision would first replace the toy interaction probability with energy-dependent neutrino–water channels and realistic detector efficiency; integrate full muon decay kinematics and ring geometry; estimate muon yield per unit primary-beam power; and rerun blind synchronization and packet error tests with physically derived event rates. The current project is intentionally closed at the feasibility stage.

Documentation note

This document reconstructs the sequence of research decisions from the project discussion and existing artifacts. It is not a laboratory notebook with independently time-stamped experimental records. Numerical values are rounded and explicitly conditional. The accompanying PDF feasibility report provides the compact final conclusion.